

A Two-Period Directed-Search Model with Wealth Heterogeneity

Equilibrium Definition and Numerical Solution

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1 Environment

This note studies a simplified version of the two-period model in Repele (2026). The simplification removes heterogeneity in job type: all jobs have the same productivity, separation probability, and vacancy-posting cost. Workers differ only in initial liquid wealth. The remaining source of heterogeneity across submarkets is the posted wage, which is associated in equilibrium with a different market tightness and hence a different job-finding probability.

There are two periods, $t = 1, 2$. A continuum of workers is indexed by initial liquid wealth

$$a_1 \sim G(a_1), \quad a_1 \in [\underline{a}_1, \bar{a}_1].$$

Workers have expected utility

$$u(c_1) + \beta \mathbb{E}[u(c_2)],$$

where $u'(c) > 0$ and $u''(c) < 0$. They can save in a risk-free asset with gross return R and face the borrowing constraint

$$a_2 \geq \underline{a}.$$

All workers start period 1 unemployed. If a worker is unemployed in period 2, she receives income b .

All jobs have common productivity $\bar{y}$, common separation probability δ , and common vacancy-posting cost ϕ . Define the retention probability

$$\Delta \equiv 1 - \delta.$$

A submarket is indexed by its posted wage w . Let $n(w)$ denote the mass of workers applying to submarket w and $v(w)$ the mass of vacancies. Market tightness is

$$\theta(w) = \frac{v(w)}{n(w)}$$

for an active submarket. A worker finds a job with probability $p(\theta)$, where $p'(\theta) > 0$, while a vacancy is filled with probability

$$q(\theta) = \frac{p(\theta)}{\theta}, \quad q'(\theta) < 0.$$

The period-2 values are

$$E_2(a_2, w) = u(Ra_2 + w), \quad U_2(a_2) = u(Ra_2 + b),$$

and the value of a surviving filled job is

$$J(w) = \bar{y} - w.$$

2 Equilibrium

A directed-search equilibrium is a collection

$$\mathcal{E} = \{a_2^*(a_1), w^*(a_1), \theta(w), n(w), v(w)\}$$

such that the following conditions hold.

- 1. Worker optimization.** Taking the tightness schedule $\theta(w)$ as given, every worker with initial wealth a_1 solves

$$\begin{aligned} (a_2^*(a_1), w^*(a_1)) \in \arg \max_{a_2 \geq \underline{a}, w} & u(Ra_1 - a_2) \\ & + \beta p(\theta(w))\Delta u(Ra_2 + w) \\ & + \beta [1 - p(\theta(w))\Delta] u(Ra_2 + b). \end{aligned} \tag{W}$$

The probability that the worker is employed in period 2 is $p(\theta(w))\Delta$: the worker must first match and the newly created match must then survive.

- 2. Firm free entry.** For every potential wage submarket w , the value of a vacancy is

$$V(w) = -\phi + \beta \Delta q(\theta(w))(\bar{y} - w).$$

Free entry and complementary slackness require

$$\begin{aligned} \phi & \geq \beta \Delta q(\theta(w))(\bar{y} - w), \quad \theta(w) \geq 0, \\ \theta(w) [\phi - \beta \Delta q(\theta(w))(\bar{y} - w)] & = 0. \end{aligned} \tag{FE}$$

Thus, whenever $\theta(w) > 0$,

$$\phi = \beta \Delta q(\theta(w))(\bar{y} - w).$$

- 3. Worker-flow consistency.** Workers' optimal search choices determine the mass of applicants to each submarket:

$$n(w) = \int \mathbf{1}\{w^*(a_1) = w\} dG(a_1). \tag{C1}$$

For a discrete wealth approximation with masses μ_i , this becomes

$$n_j = \sum_i \mu_i \mathbf{1}\{w_i^* = w_j\}.$$

- 4. Vacancy/tightness consistency.** For every active submarket,

$$v(w) = \theta(w)n(w). \tag{C2}$$

Hence $\theta(w) = v(w)/n(w)$ whenever $n(w) > 0$. If no worker chooses a submarket, set $n(w) = v(w) = 0$; the zero-profit schedule $\theta(w)$ can still be viewed as the tightness that would prevail were that submarket active.

5. Matching consistency. Job-finding and vacancy-filling probabilities are generated by the same tightness:

$$p_w = p(\theta(w)), \quad q_w = q(\theta(w)) = \frac{p(\theta(w))}{\theta(w)}. \quad (\text{C3})$$

There is no asset-market-clearing condition because R is taken as exogenous.

3 Why there is no fixed point over the tightness schedule

In this simplified economy, conditional on a wage w , free entry determines $\theta(w)$ without reference to the mass or composition of workers who choose that submarket. The firm's expected profit depends on w and θ , but not on the absolute numbers of workers and vacancies. Constant returns to scale in matching imply that only their ratio matters.

For an active submarket,

$$q(\theta(w)) = \frac{\phi}{\beta \Delta(\bar{y} - w)}. \quad (1)$$

Since $q(\theta)$ is decreasing, this determines a unique $\theta(w)$ whenever the right-hand side lies in the range of q . Workers then choose among the resulting submarkets. Their choices determine $n(w)$, and vacancy entry adjusts mechanically to

$$v(w) = \theta(w)n(w).$$

Thus the computational sequence is

$$w \longrightarrow \theta(w) \longrightarrow \text{worker choices} \longrightarrow n(w) \longrightarrow v(w).$$

There is no need to guess $\theta(w)$, solve workers' choices, and iterate back to a new $\theta(w)$. A genuine fixed point would arise if firm profitability depended on the endogenous composition of applicants, for example if worker productivity differed by wealth or another worker type.

4 A convenient matching-function parameterization

Following the functional form used by Repele (2026, Appendix B1), consider the constant-returns matching function

$$M(n, v) = nv (n^\gamma + v^\gamma)^{-1/\gamma}, \quad \gamma > 0.$$

With $\theta = v/n$, the worker job-finding and firm vacancy-filling probabilities are

$$p(\theta) = \theta (1 + \theta^\gamma)^{-1/\gamma},$$

$$q(\theta) = (1 + \theta^\gamma)^{-1/\gamma}.$$

These probabilities are bounded between zero and one and satisfy $q(\theta) = p(\theta)/\theta$.

For this matching function, the free-entry schedule can be computed without a root finder. Define

$$q^{FE}(w) \equiv \frac{\phi}{\beta \Delta(\bar{y} - w)}.$$

For $0 < q^{FE}(w) < 1$,

$$\theta(w) = \left([q^{FE}(w)]^{-\gamma} - 1 \right)^{1/\gamma}, \quad (2)$$

and then

$$p(w) = p(\theta(w)).$$

If $q^{FE}(w) \geq 1$, the vacancy cannot cover its posting cost even when it is filled with probability one, so that submarket is inactive and $\theta(w) = 0$.

Repele's quantitative model uses $\gamma = 0.38$, which implies an average elasticity of job finding with respect to tightness in the range targeted in her calibration. For the simplified two-period exercise, $\gamma = 0.38$ is therefore a natural benchmark, although it is not necessary for the theoretical results.

5 Numerical parameterization and grids

A useful way to parameterize the toy model is to normalize productivity and calibrate the vacancy cost to a reference job-finding rate rather than choosing ϕ arbitrarily.

Preferences and common job parameters

A convenient utility specification is CRRA:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad \sigma > 0, \sigma \neq 1,$$

with $u(c) = \log c$ when $\sigma = 1$. Repele's quantitative model uses $\sigma = 2$, which is a useful benchmark here as well.

Normalize

$$\bar{y} = 1.$$

Choose R and β consistently with the interpretation of one model period. Choose the common separation probability δ from the desired time frequency, and set $\Delta = 1 - \delta$. A simple benefit parameterization is

$$b = \rho_b \bar{y},$$

where ρ_b is a replacement-rate parameter.

Calibrating the vacancy cost

Choose a reference wage $\bar{w}$ and a target job-finding probability $\bar{p} \in (0, 1)$. Under the matching function above,

$$\bar{\theta} = \left(\frac{\bar{p}^\gamma}{1 - \bar{p}^\gamma} \right)^{1/\gamma},$$

and

$$\bar{q} = q(\bar{\theta}) = (1 - \bar{p}^\gamma)^{1/\gamma}.$$

Then set

$$\phi = \beta \Delta \bar{q} (\bar{y} - \bar{w}). \quad (3)$$

This guarantees that $(\bar{w}, \bar{p})$ lies on the free-entry schedule.

Wage grid

For an active submarket the free-entry equation requires

$$q^{FE}(w) \leq 1,$$

which implies

$$w \leq \bar{y} - \frac{\phi}{\beta\Delta}.$$

Moreover, with no nonpecuniary value of employment, wages weakly below b are dominated for workers. A practical wage grid is therefore

$$w \in [w_{\min}, w_{\max}],$$

with

$$\begin{aligned} w_{\min} &= b + \varepsilon_w, \\ w_{\max} &= \bar{y} - \frac{\phi}{\beta\Delta} - \varepsilon_w, \end{aligned}$$

where $\varepsilon_w > 0$ is small. The parameterization should satisfy $w_{\min} < w_{\max}$. A grid of roughly $N_w = 200$ – 500 points is more than sufficient for this two-period problem.

A uniform wage grid is transparent. Numerically, however, $\theta(w)$ can change quickly near the upper wage bound. An alternative is to use a grid that is uniform in p or θ and recover the corresponding wage from

$$w(\theta) = \bar{y} - \frac{\phi}{\beta\Delta q(\theta)}.$$

Wealth grid

With a borrowing limit $\underline{a} = 0$, choose

$$a_1 \in [0, a_{\max}].$$

It is useful to place more grid points near zero because wealth has the largest effect on search choices close to the borrowing constraint. For example, with $i = 1, \dots, N_a$,

$$a_{1,i} = a_{\max} \left(\frac{i-1}{N_a-1} \right)^\kappa, \quad \kappa > 1.$$

Values such as $\kappa = 2$ and $N_a = 200$ are adequate for a simple numerical illustration. Attach probability weights μ_i either from a chosen parametric distribution or directly from empirical liquid-wealth data.

6 Solution algorithm

For a discrete numerical implementation, the following algorithm computes the equilibrium directly.

Step 1. Choose grids $\{w_j\}_{j=1}^{N_w}$ and $\{a_{1,i}\}_{i=1}^{N_a}$ and probability weights $\{\mu_i\}$ for the wealth distribution.

Step 2. For every wage w_j , compute

$$q_j^{FE} = \frac{\phi}{\beta \Delta(\bar{y} - w_j)}.$$

If $0 < q_j^{FE} < 1$, compute θ_j from equation (2), and then $p_j = p(\theta_j)$. If $q_j^{FE} \geq 1$, set the submarket to inactive.

Step 3. For each pair $(a_{1,i}, w_j)$, solve the worker's one-dimensional savings problem

$$\max_{a_2 \geq \underline{a}} \{u(Ra_{1,i} - a_2) + \beta p_j \Delta u(Ra_2 + w_j) + \beta(1 - p_j \Delta)u(Ra_2 + b)\}. \quad (4)$$

With positive-consumption utility and $\underline{a} \geq 0$, a convenient feasible interval is

$$a_2 \in [\underline{a}, Ra_{1,i} - \varepsilon_c].$$

The problem can be solved by a scalar optimizer or, when the solution is interior, by the Euler equation.

Step 4. For each wealth type $a_{1,i}$, compare the optimized value across wage submarkets and choose

$$j^*(i) \in \arg \max_j \tilde{U}(a_{1,i}; w_j).$$

Set

$$w_i^* = w_{j^*(i)}, \quad \theta_i^* = \theta_{j^*(i)},$$

and retain the corresponding optimal savings $a_{2,i}^*$.

Step 5. Count the mass of workers applying to each submarket:

$$n_j = \sum_i \mu_i \mathbf{1}\{j^*(i) = j\}.$$

This is the mass of period-1 searchers in submarket j , not an unemployment rate.

Step 6. For every active submarket, create the mass of vacancies

$$v_j = \theta_j n_j.$$

If $n_j = 0$, set $v_j = 0$.

Step 7. Verify equilibrium numerically. For every active submarket check

$$|\phi - \beta \Delta q(\theta_j)(\bar{y} - w_j)| < \text{tol},$$

and

$$\left| \theta_j - \frac{v_j}{n_j} \right| < \text{tol}.$$

Also verify that each worker's chosen wage attains the maximum of the worker problem over the wage grid.

The output of the algorithm is the wealth-dependent policy map

$$a_1 \mapsto (a_2^*(a_1), w^*(a_1), \theta^*(a_1), p^*(a_1)),$$

together with the equilibrium masses $n(w)$ and $v(w)$. The main economic object of interest is how wealth changes the worker's willingness to trade a higher wage for a lower job-finding probability.

7 Reference

References

- [1] Repele, Amalia (2026), *Wealth Sorting and Cyclical Employment Risk*, Job Market Paper, International Institute for Economic Studies, Stockholm University, July 2026.